

Second-Order Linear Homogeneous Equations

ANSWER KEY



The auxiliary equation

- 1 Write the auxiliary (characteristic) equation for $y'' - 5y' + 6y = 0$.

$$r^2 - 5r + 6 = 0.$$

- 2 Write and simplify the auxiliary equation for $2y'' + 4y' - 6y = 0$.

$$2r^2 + 4r - 6 = 0, \text{ which simplifies to } r^2 + 2r - 3 = 0.$$

Distinct real roots

- 3 Solve $y'' - 5y' + 6y = 0$.

$$r^2 - 5r + 6 = (r - 2)(r - 3) = 0 \Rightarrow r = 2, 3. \text{ General solution: } y = C_1 e^{2x} + C_2 e^{3x}.$$

- 4 Solve $y'' + 2y' - 3y = 0$, $y(0) = 1$, $y'(0) = 1$.

$$r^2 + 2r - 3 = (r + 3)(r - 1) = 0 \Rightarrow r = -3, 1. \text{ So } y = C_1 e^{-3x} + C_2 e^x. \text{ From } y(0) = 1: C_1 + C_2 = 1. \text{ From } y'(0) = 1: -3C_1 + C_2 = 1. \text{ Subtracting gives } 4C_1 = 0, \text{ so } C_1 = 0, C_2 = 1. \text{ Thus } y = e^x.$$

Repeated roots

- 5 Solve $y'' - 6y' + 9y = 0$.

$$r^2 - 6r + 9 = (r - 3)^2 = 0 \Rightarrow r = 3 \text{ (repeated)}. \text{ General solution: } y = (C_1 + C_2 x)e^{3x}.$$

- 6 Solve $y'' + 4y' + 4y = 0$, $y(0) = 2$, $y'(0) = -1$.

$$r^2 + 4r + 4 = (r + 2)^2 = 0 \Rightarrow r = -2. \text{ So } y = (C_1 + C_2 x)e^{-2x}. \text{ From } y(0) = 2: C_1 = 2. \text{ Differentiating, } y' = C_2 e^{-2x} - 2(C_1 + C_2 x)e^{-2x}; y'(0) = C_2 - 2C_1 = -1 \Rightarrow C_2 = 2(2) - 1 = 3. \text{ Thus } y = (2 + 3x)e^{-2x}.$$

Complex roots

- 7 Solve $y'' + 4y = 0$.

$$r^2 + 4 = 0 \Rightarrow r = \pm 2i. \text{ General solution: } y = C_1 \cos 2x + C_2 \sin 2x.$$

- 8 Solve $y'' - 2y' + 5y = 0$.

$$r^2 - 2r + 5 = 0 \Rightarrow r = \frac{2 \pm \sqrt{4 - 20}}{2} = 1 \pm 2i. \text{ General solution: } y = e^x (C_1 \cos 2x + C_2 \sin 2x).$$

Application: a damped spring

- 9 A mass–spring–damper system satisfies $y'' + 4y' + 5y = 0$. Find the general solution and classify the damping (over/critically/under-damped).

$r^2 + 4r + 5 = 0$, discriminant $= 16 - 20 = -4 < 0$, so $r = -2 \pm i$. General solution:

$y = e^{-2x}(C_1 \cos x + C_2 \sin x)$ — this is under-damped (complex roots with negative real part give oscillatory decay).

- 10 Solve $y'' + 6y' + 9y = 0$, $y(0) = 1$, $y'(0) = 0$, and classify the damping.

$r^2 + 6r + 9 = (r + 3)^2 = 0 \Rightarrow r = -3$ (repeated) — critically damped. So $y = (C_1 + C_2x)e^{-3x}$. From

$y(0) = 1$: $C_1 = 1$. $y' = C_2e^{-3x} - 3(C_1 + C_2x)e^{-3x}$; $y'(0) = C_2 - 3C_1 = 0 \Rightarrow C_2 = 3$. Thus $y = (1 + 3x)e^{-3x}$.

Further practice: distinct real roots

- 11 Solve $y'' - y' - 6y = 0$.

$r^2 - r - 6 = (r - 3)(r + 2) = 0 \Rightarrow r = 3, -2$. General solution: $y = C_1e^{3x} + C_2e^{-2x}$.

- 12 Solve $y'' - 4y = 0$.

$r^2 - 4 = 0 \Rightarrow r = \pm 2$. General solution: $y = C_1e^{2x} + C_2e^{-2x}$.

Further practice: repeated roots

- 13 Solve $y'' - 8y' + 16y = 0$.

$r^2 - 8r + 16 = (r - 4)^2 = 0 \Rightarrow r = 4$ (repeated). General solution: $y = (C_1 + C_2x)e^{4x}$.

- 14 Solve $y'' + 10y' + 25y = 0$, $y(0) = 0$, $y'(0) = 3$.

$r^2 + 10r + 25 = (r + 5)^2 = 0 \Rightarrow r = -5$. So $y = (C_1 + C_2x)e^{-5x}$. From $y(0) = 0$: $C_1 = 0$.

$y' = C_2e^{-5x} - 5(C_1 + C_2x)e^{-5x}$; $y'(0) = C_2 - 5C_1 = 3 \Rightarrow C_2 = 3$. Thus $y = 3xe^{-5x}$.

Further practice: complex roots

- 15 Solve $y'' + 9y = 0$.

$r^2 + 9 = 0 \Rightarrow r = \pm 3i$. General solution: $y = C_1 \cos 3x + C_2 \sin 3x$.

- 16 Solve $y'' + 2y' + 10y = 0$.

$r^2 + 2r + 10 = 0 \Rightarrow r = \frac{-2 \pm \sqrt{4 - 40}}{2} = -1 \pm 3i$. General solution: $y = e^{-x}(C_1 \cos 3x + C_2 \sin 3x)$.

Reading the discriminant without fully solving

- 17 Without fully solving, state whether $y'' + 6y' + 9y = 0$ has distinct real, repeated, or complex roots.

$r^2 + 6r + 9$: discriminant $= 36 - 36 = 0$, so the roots are repeated.

- 18 Without fully solving, classify the roots of $y'' + y' + y = 0$.

Discriminant $= 1 - 4 = -3 < 0$, so the roots are complex.

Application: an undamped spring

- 19** An undamped mass-spring system satisfies $y'' + 16y = 0$. Find the general solution and the angular frequency of oscillation.

$r^2 + 16 = 0 \Rightarrow r = \pm 4i$. General solution: $y = C_1 \cos 4x + C_2 \sin 4x$. Angular frequency: $\omega = 4$.

- 20** The same spring is released from rest at displacement 2: $y(0) = 2$, $y'(0) = 0$. Find the particular solution.
From $y(0) = 2$: $C_1 = 2$. $y' = -8 \sin 4x + 4C_2 \cos 4x$; $y'(0) = 4C_2 = 0 \Rightarrow C_2 = 0$. Thus $y = 2 \cos 4x$.

One more round: mixed root types

- 21** Solve $y'' - 9y = 0$, $y(0) = 4$, $y'(0) = 0$.

$r^2 - 9 = 0 \Rightarrow r = \pm 3$. So $y = C_1 e^{3x} + C_2 e^{-3x}$. From $y(0) = 4$: $C_1 + C_2 = 4$. $y' = 3C_1 e^{3x} - 3C_2 e^{-3x}$; $y'(0) = 3C_1 - 3C_2 = 0 \Rightarrow C_1 = C_2$. So $C_1 = C_2 = 2$: $y = 2e^{3x} + 2e^{-3x}$.

- 22** Solve $y'' + 2y' + 2y = 0$.

$r^2 + 2r + 2 = 0 \Rightarrow r = \frac{-2 \pm \sqrt{4-8}}{2} = -1 \pm i$. General solution: $y = e^{-x}(C_1 \cos x + C_2 \sin x)$.

One final pair

- 23** Solve $y'' - y = 0$.

$r^2 - 1 = 0 \Rightarrow r = \pm 1$. General solution: $y = C_1 e^x + C_2 e^{-x}$.

- 24** Solve $y'' + 25y = 0$.

$r^2 + 25 = 0 \Rightarrow r = \pm 5i$. General solution: $y = C_1 \cos 5x + C_2 \sin 5x$.

- 25** Solve $y'' - y' - 2y = 0$, $y(0) = 0$, $y'(0) = 3$.

$r^2 - r - 2 = (r-2)(r+1) = 0 \Rightarrow r = 2, -1$. So $y = C_1 e^{2x} + C_2 e^{-x}$. From $y(0) = 0$: $C_1 + C_2 = 0$.
 $y' = 2C_1 e^{2x} - C_2 e^{-x}$; $y'(0) = 2C_1 - C_2 = 3$. Adding to $C_2 = -C_1$: $3C_1 = 3 \Rightarrow C_1 = 1$, $C_2 = -1$. Thus
 $y = e^{2x} - e^{-x}$.

- 26** Solve $y'' + 2y' + y = 0$.

$r^2 + 2r + 1 = (r+1)^2 = 0 \Rightarrow r = -1$ (repeated). General solution: $y = (C_1 + C_2 x)e^{-x}$.

A closing pair with initial conditions

- 27** Solve $y'' - 4y' + 13y = 0$.

$r^2 - 4r + 13 = 0 \Rightarrow r = \frac{4 \pm \sqrt{16-52}}{2} = 2 \pm 3i$. General solution: $y = e^{2x}(C_1 \cos 3x + C_2 \sin 3x)$.

28 Solve $y'' + 3y' = 0$, $y(0) = 5$, $y'(0) = -6$.

$r^2 + 3r = r(r + 3) = 0 \Rightarrow r = 0, -3$. So $y = C_1 + C_2e^{-3x}$. From $y(0) = 5$: $C_1 + C_2 = 5$. $y' = -3C_2e^{-3x}$;
 $y'(0) = -3C_2 = -6 \Rightarrow C_2 = 2$, so $C_1 = 3$. Thus $y = 3 + 2e^{-3x}$.